AI Agent for Service Coordinator Project

Executive Summary & Objective

Vision Statement

An internal AI assistant (in the form of a chatbot similar to the one on Cummins Answers) that supports a service coordinator in daily tasks. It should know the do’s and don’ts of the service department, centralizing tribal knowledge, and enforcing operational compliance (labor limits, billing rules). Its main objective is to reduce unnecessary communication loops between clients, coordinators, and technicians. Furthermore, it could act as an on-demand co-pilot to accelerate new hire onboarding..

Success Metrics

Time-to-Productivity: Reduce the onboarding time for a new coordinator from 4 weeks to 2 weeks (to be refined).

Interruption Rate: Reduce the number of times a new coordinator has to interrupt a senior coordinator/manager for “nuance” questions by 50%.

Compliance Errors: Zero instances of technicians accidentally booked past their 14-hour daily work limit

Core Users & Pain Points

The core user will be the service coordinator. Pain points are most likely to affect new coordinators. Within the service department, there are a lot of nuances regarding how exactly the department functions. Examples include knowing which technicians are able to perform certain tasks, invoicing cash accounts, and the 14-hour working limit for a technician. Currently available training documents are sparse and only cover the basics. Most of the time, new coordinators would resort to learning on the go and ringing up senior coordinators, which reduces the overall effectiveness of the department.

Scope of the Project

In-Scope

Consolidate the existing training materials, refine them, and prepare them to feed into the AI Agent. It has to be as comprehensive as possible. However, we could scale its knowledge later on and start with what we have currently.

Build the foundation of the chatbot, test its knowledge internally once the chatbot becomes live and refine the model until it becomes dependable.

The AI will strictly provide guidance and information based on static knowledge text, rather than executing live actions.

Out-of-Scope

Voice call analysis and automatically dispatching technicians without coordinator supervision. Direct integration with FieldAware and BMS live databases.

System Integration & UI

The assistant would live in the Route Optimizer application developed by AR76F. The Optimizer should also become a central ecosystem for coordinators. The UI of the assistant should mimic the one found on Cummins Answers for consistency.

Knowledge Structure

The following would define how exactly knowledge would be stored in the assistant:

- Category A: Regulatory and Compliance (Hard Rules)
    - Data Source needed: Documented labor laws, company safety policies, and the 14-hour technician working limit.
- Category B: Financial & Job Planning Nuances (Processes):
    - Data Source needed: Step-by-step workflows for handling various tasks such as invoicing, and job booking. Includes invoicing exceptions, and other protocols.
- Category C: Skill & Capability Mapping (Logistics):
    - Data Source needed, a static matrix/spreadsheet of current technicians paired with their specific certifications, preferences, and experience.
- Category D: The tribal knowledge:
    - Data Source needed: A running log of unwritten rules, common pitfalls, and what-if scenarios gathered by the department.

Phased Rollout, Testing, & Alignment Plan

Phase 1: Knowledge Gathering and Structuring

Consolidate existing training material, gather unwritten department rules. Format all info into clear, structured Markdown documents.

Phase 2: Alpha Testing and Guardrail Tuning

Development of the chatbot itself then embed it into the Route Optimizer in a closed testing environment. Stress-test by various service coordinators. Adjust the AI responses as needed. The goal is to achieve a 90% accuracy rate on core department rules.

Phase 3: Pilot Launch

Deploy the chatbot to a small pilot group and track the interruption rate (is the new hire using the chatbot or still stopping senior coordinators/manager)

Phase 4: Full Deployment and Future Roadmap

Deploy the tool to the entire service department. Review Power BI data to evaluate scheduling efficiencies and monitor compliance errors.

Future scope includes developing a voice call analysis feature where the screening process of a service call would be entirely handled by AI. Live API integrations with FieldAware and BMS should be on the horizon as well, barring administrative obstacles.